

Relation Algebra

§9. $\frac{R}{S}$

$$R/S \triangleq (R/S) \cap (S/R)^{\circ}$$

what R sends x to and nothing else is what S sends y to. In Rel x and y has the same image: $\forall p. (x R p \iff y S p)$

(9a)

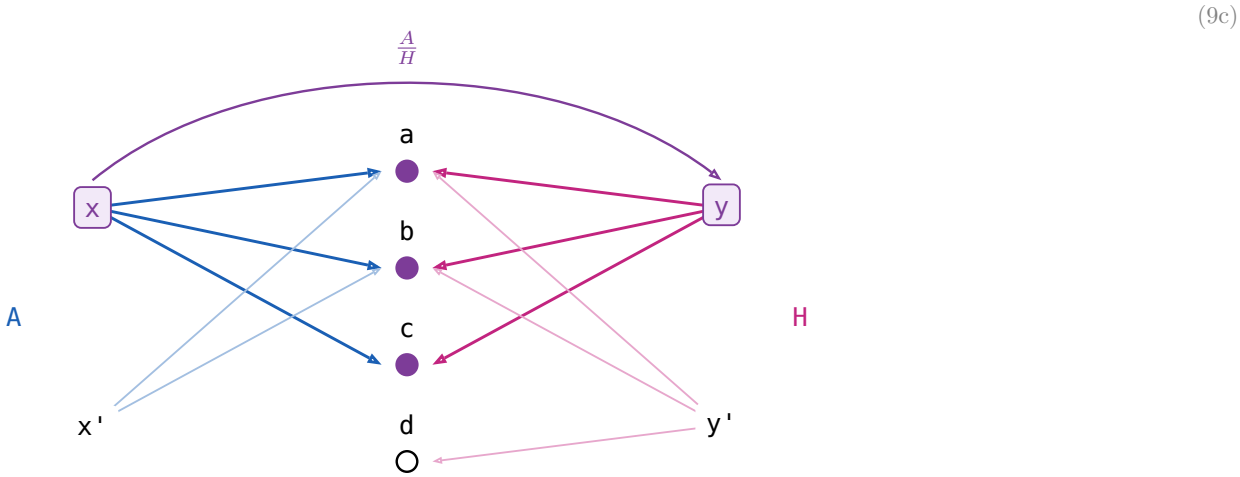
$$x (A/H) y \text{ --- } x \text{ admires everyone } y \text{ hates}$$

$$x ((H/A)^{\circ}) y \text{ --- } x \text{ admires only people } y \text{ hates}$$

$$x ((A/H) \cap (H/A)^{\circ}) y \text{ --- } x \text{ admires only and all whom } y \text{ hates}$$

(9b)

x admires exactly whom y hates: $/$ is **all of**, $\frac{R}{S}$ is **only all of**.



law	the reading
$X \in R / {}_s S \iff X S \in R \wedge X {}^\circ R \in S$ X sits inside the matching exactly when it carries S back inside R and, reversed, carries R back inside S .	X may pair x with y only when x admires exactly whom y hates. Both halves must typecheck, so the operation is partial .
$(R / {}_s S) {}^\circ = S / {}_s R$ reversing the matching swaps the two relations.	Matching is symmetric.
$(R / {}_s S) (S / {}_s W) \in R / {}_s W$ a match from R to S followed by one from S to W is a match from R to W .	And transitive.
$(R / {}_s S) S \in R$ following the matching by S lands inside R .	$(\exists y. x (\frac{R}{S}) y \wedge y S p) \rightarrow x R p$ x only admires whom y hates $\frac{R}{S} S = \text{Dom}(\frac{R}{S}) R$
$(R / {}_s R) R = R$ matching R against itself and then following R gives R back.	$(\exists y. x (\frac{R}{R}) y \wedge y R p) \iff x R p$ x and y admire the same people $y=x$ always qualifies ($1 \in R \% R$ below)
$1 \in R / {}_s R$ every x matches itself.	$x (\frac{R}{R}) y$ if x and y admires the same peoples.
$(R / {}_s R) (R / {}_s R) = R / {}_s R$ matching twice matches no more pairs than matching once.	So the relation admires the same people is an equivalence relation.
$\text{Symmetric}(X) \implies X \in R / {}_s R \iff X R \in R$ a symmetric X sits inside the matching exactly when following it by R adds nothing to R .	The largest symmetric arrow that leaves R alone.
$\text{simplePart}(R) \triangleq R / {}_s 1$ matching R against the identity keeps the x whose image is a single point.	The people who admire exactly one person and nobody else. It equals R only when R is simple, unlike $R / 1 = R$.
$\text{dom}(R / {}_s S) = 1 n (R / S) (S / R)$ the x that match something are those the two one-sided divisions take round the loop back to x .	Its domain is the domain of simplicity of R .